

# DSA (Abdul Bari) - Day 43

45-day plan | YouTube playlist course

## Dynamic Programming - Intro

Solve overlapping subproblems once.

### Key concepts

- Optimal substructure + overlapping subproblems.
- Memoisation (top-down) vs tabulation (bottom-up).
- Store results to avoid recomputation.

### Key idea / example

fib with memo turns  $O(2^n)$  into  $O(n)$ .

### Today's tasks (1.5h)

- ☐ Memoise fibonacci
- ☐ Convert to tabulation
- ☐ Spot overlapping subproblems

### Resources

Playlist: [youtube.com/playlist?list=PLsr8vTgyLdy\\_YndxNcI4WkH5Vorj5qvrV](https://youtube.com/playlist?list=PLsr8vTgyLdy_YndxNcI4WkH5Vorj5qvrV)

(Clickable playlist link is in the app: Goals > this day > Study block.)